SPEAKING SCRIPT

Backend

Ayapong Albert Yeboah

Backend lead

Duration	about 2 minutes 15 seconds
Slot in the video	Fifth (after Benjamin)
On camera	Face, then VS Code on backend routes / health endpoint

How to use this script

Read the spoken words naturally. Do not read lines in [SQUARE BRACKETS]; those are camera or screen cues. If you run long, skip a paragraph marked OPTIONAL. Aim for the duration on the cover. The full team video should land between 8 and 15 minutes.

What the API is

[SHOW: terminal curl to /api/health, or Postman. Then a quick look at backend/src/index.js.]

I am Albert. I lead the backend. The server is Node and Express. Every screen you just saw talks to REST routes: auth, products, categories, sales, reports, backup, and sync. We issue JWTs after login. The password is hashed. If the username is wrong we still do a dummy hash so we do not leak whether an account exists.

Two modes, one interface. Standalone mock needs no Postgres, so the team could demo on day one. Connected mode is real Postgres on Render. Routes do not care which one is underneath. That was an architecture decision: we would not block frontend on the database being ready.

Sales and profit - the important part

[OPTIONAL: show a sale JSON with unit_price and unit_cost.]

When you record a sale, the API does not trust the browser's prices. It locks the product, checks quantity, and stamps two numbers on the line: the current shelf price, and the cost of the stock being sold. Cost comes from FIFO layers, oldest first. If leftover old stock is still on the shelf, we do not pretend it cost the new supplier price. Profit reports then add those stamped lines. Change the product tomorrow; yesterday's sale does not rewrite.

Sync is the same create-sale path for the offline queue, with a whitelist on product fields so a client cannot inject a column name. Backup snapshots include those layers so a restore does not flatten history.

Security, in one breath

Helmet headers, CORS allow-list for Vercel, rate limits on login, small JSON bodies, parameterized SQL, owner-only backup, and no secrets in git. Finance-adjacent data does not get a student demo password in production. Change owner-one-two-three-four.

Deploy

API and Postgres live on Render. Frontend on Vercel. Health check keeps the story honest: status ok, mode postgres or mock. Francis and Raymond will show the tables that make those stamps possible.